Initial set up: If MatLab 2015a is not already in the applications folder, download it from the Campus Software Library and install it. Copy the "Sphinctalyzer10312015" folder into the MatLab folder in the documents folder (folder is in the shared folder on the iMac, if the MatLab folder isn't there, open the program and then it should appear). Now the "Sphinctalyzer10312015" folder should show up on the left-hand portion of the MatLab screen that shows the current folder. Double click on the programs you want to run and they will appear in the editor portion of the screen for you to use.

To bulk process multiple samples at a time:

In FIJI:

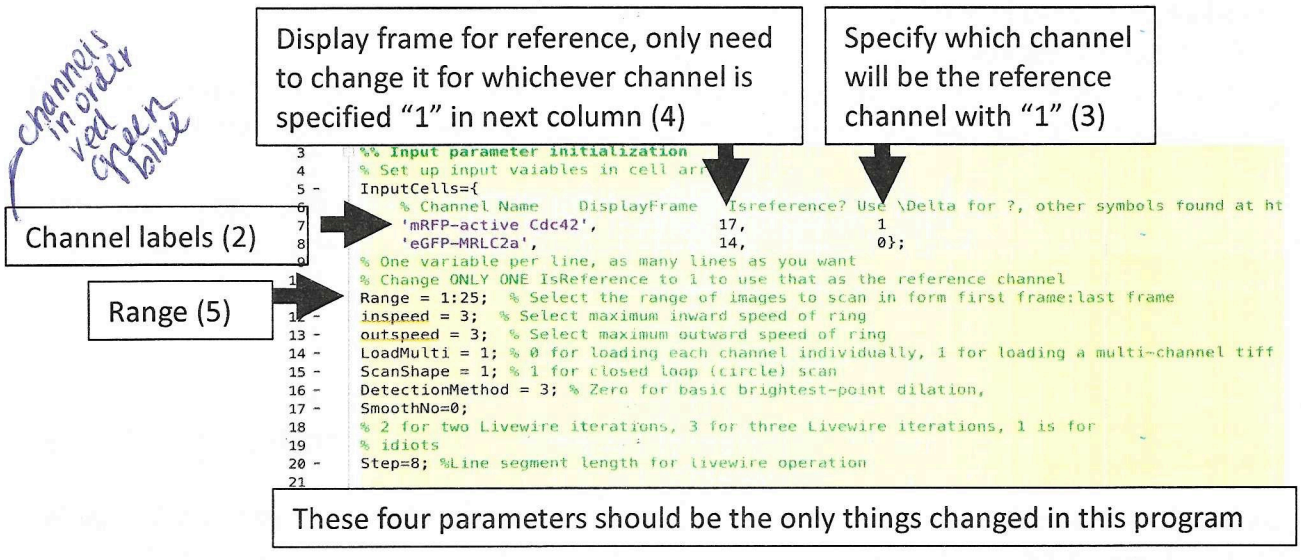
1) Create projections and make sure they are saved as composite images (you can load multi-channel movies into the program)
2) Determine time point at which you will manually draw the brightest path in MatLab (look for the time point which has the most well defined zones) and determine number of frames in movie for all movies you want to Sphinctalyze

In your data folder:

1) Create a folder labelled "sphinc" (has to be lowercase s) within your data folder to store sphinctalyzed files

In Matlab:

1) Open SphinctalyzerInitializer (the green text in Matlab gives instructions as to what each variable means and should never be changed; also don't change anything you don't understand)
2) Label channels appropriately (purple text). If using three channels, make sure there are three lines in the InputCells section
3) Determine which channel will be used as a reference and change the IsReference value to 1 for that channel (all other channels should be 0)
4) Change Display Frame for reference channel to the frame determined in FIJI (this is the time point used to manually draw the brightest path). You do not need to change it for the channels that are not the reference
5) Change the range to include all time points in your movie determined in FIJI



6) Run SphinctalyzerInitializer (green arrow)
7) Choose tiff to sphinctalyze
8) Name file to save (must include "sphinc" in file name, do not include any punctuation/special characters in the file name) and save it to a folder with the name "sphinc" in it
    a. Any text before "sphinc" in the file name is how it will be labeled in the data spreadsheet

9) Manually draw the brightest path, making sure not to completely close the loop because the program will do this for you (if you did not draw the path well, right click and delete it, then draw it again)
10) Double click inside the brightest path shape to run the SphinctalyzerInitializer
11) Repeat steps 2-10 for all movies (this process creates a file with the brightest path calculated and refined for only the single time point specified for each movie)
12) Once all movies are initialized, open SphincRescue (this program will calculate and refine the brightest path for all time points based on the SphinctalyzerInitializer information)
13) Run SphincRescue (green arrow) and choose the folder(s) you'd like to Sphinctalyze. Make sure to always select a folder one step above the folder you want to run (ie don't run the "sphinc" folder, run the folder containing the "sphinc" folder). For example, if you SphincRescue nested folders, all folders with the word "sphinc" in them will be looked at and any files with the word "sphinc" contained within any of the "sphinc" folders will be run, which allows you to run multiple data sets at once.
14) Let SphincRescue run. It may take a few hours depending on the number of samples run. Make sure to change your computer settings so that it will not go to sleep while SphincRescue is running. A new folder/folders will be created labelled SphincRedo which contains each movie sphinctalyzed for all time points according to the specifications you determined.
15) To get numbers/brightest path scan movies from the sphinctalyzed data, open SphinctalyzerMoviePlotterMulti. *If you went straight thru from initializing and rescuing, you need to close MatLab and re-open it to reset the file directory. (See #3 in Tips and Tricks).
16) You may change lines 39-59 of this program. The most common changes are to the method of normalization, microns per pixel, and time per frame. Alter these as needed.
17) Run SphinctalyzerMoviePlotterMulti (green arrow) and choose the folder that contains your "SphincRedo" folder to run. This program will also work with nested folders to run multiple experiments at a time (assuming the settings such as microns per pixel and time per frame are the same).
18) The output will be an Excel file with data for all time points of every sample for PeakIntensity, PeakLocation, Halfwidth, AUC, and percent change. You will also get a brightest path scan movie saved as an AVI for every sample.

Sphinctalyzing individual movies in MatLab

1) Open Sphinctalyzer program.
2) Change labels, reference channel, display frame, and range (directions 2-5 in bulk processing section)
3) Run Sphinctalyzer (directions 6-10 in bulk processing section). For one movie, it should take approximately 2-5 minutes.
4) To create a brightest path scan movie, open SphinctalyzerMoviePlotterZeroNorm, change parameters accordingly (usually just normalization method, microns per pixel, and time per frame). Then run the program and point it to your Sphinctalyzer output that you ran in step 3.

To create a brightest path scan at a single time point

1) Open SphinctalyzerMultiFramePlotter
2) Adjust normalization method, microns per pixel, time per frame, etc as appropriate (lines 3-21 of the code)
3) Use the CropRange feature in line 9 to specify which time points you'd like a scan of (if you'd only like one time point you still need to specify more than one time point or the program won't run).
4) Run SphinctalyzerMultiFramePlotter

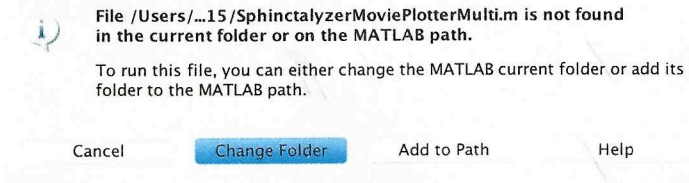
Creating an outline movie in MatLab (overlays the brightest path on top of the actual image)

1) To test how well your brightest path was calculated by the Sphinctalyzer, run SphincOutlineMovie

2) Choose ChannelToPlot in line 1
3) Choose outline color in line 2
4) Run SphincOutlineMovie and point the program to an individual sphinctalyzed file.

Tips and tricks:

1) Do NOT modify any parameters or code that you do not understand!
2) To save time, change the current folder by clicking "browse for folder" button and choosing the folder with your data in it, and when running a program, choose the "add to path" button and you will automatically be redirected to this folder instead of the MatLab folder.
3) If the SphinctalyzerMoviePlotterMulti is having trouble, close MatLab and reopen it. In order for MatLab to create the Excel file correctly, do not use the "add to path" feature but choose "change folder" and then find your data.



4) Make sure your SphincRedo folder actually says "SphincRedo" in the file name. Depending on how you labelled your "sphinc" folder there may be words/spaces between "sphinc" and "redo" and the SphinctalyzerMoviePlotterMulti will not recognize this naming scheme.
5) Do not use any sort of punctuation/special characters in any folder or file names.